



Krushang Prajapati

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PROFILE OVERVIEW

Second-year Electronics and Communication Engineering student (Sem 4) with hands-on experience in embedded systems, IoT development, and digital electronics. Skilled in working with ESP32, Firebase-based dashboards, circuit simulation using LTspice, and Linux environments.

Interested in hardware–software integration, AI-based vision systems, FPGA/VLSI design, and cybersecurity projects. Passionate about building practical systems that combine electronics, cloud platforms, and intelligent automation.

EDUCATION

B.Tech — Electronics and Communication Engineering (Pursuing)

Sarvajanik College of Engineering & Technology, Sarvajanik University, Surat

2024 – 2028

SPI: 7.3/ 10 (Sem 3)

HSC — Class XII, Science (PCM)

BhagwanMahavir School, Surat | CBSE

2024

64%

SSC — Class X

B.S Patel Primary School, Bilimora | CBSE

2022

79%

SKILLS

Embedded & IoT: ESP32, Arduino, Sensor Integration, IoT System Design

Electronics: Analog & Digital Electronics, Op-Amps, Diodes, Transistors

Circuit Simulation: LTspice, MATLAB Simulink

Programming: Python (Basic), C (Basic), MATLAB

Cloud & Backend: Firebase Realtime Database, IoT Dashboard Development

Systems: Linux (Arch Linux), Command Line Tools

Machine Learning: Basic ML Concepts, AI-based Motion Detection

Hardware: ESP32-CAM, Sensors, Relay Modules

Soft Skills: Critical Thinking, Leadership, Communication, Team Collaboration

ACADEMIC PROJECTS

FPGA-Based Digital Design Implementations

Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA

- Implemented Sequence Detector (Moore & Mealy FSM), Sequence Generator, arithmetic units (Adder / Subtractor), Flip-Flops, MUX, and DSD modules in Verilog HDL — validated on FPGA.

ESP32-CAM AI Motion Detection & Monitoring System

Academic Project | ESP 32 - CAM | FireBase | AI Motion Detection

- Developed a smart surveillance system using ESP32-CAM integrated with an AI-based motion detection model. Detected movement through camera input and sent alerts/data to a Firebase realtime dashboard for remote monitoring.

XOR-Based Symmetric Encryption Engine

Academic Project | Verilog HDL | FPGA

- Implemented parameterised encryption / decryption hardware with configurable key width and key-locked decryption logic — validated on FPGA.

Motion Detection System using Computer Vision

Academic Project | Python | OpenCV | Background Subtraction (MOG2)

- Implemented a motion detection system using computer vision techniques. Used background subtraction to detect movement and trigger events for monitoring applications.

POSITIONS OF RESPONSIBILITY

Google Student Ambassador

Aug 2025 – Jan 2026

Hybrid Mode

- Explored Gemini AI capabilities; conducted sessions training college students on using AI tools for academic and professional workflows.

IEEE Student Member | T&P Cell Volunteer

2024 – 2025

ACHIEVEMENTS

- Won Chess Tournament , SCET, Surat
- Participated in Smart India Hackathon (SIH) 2025 — SCET
- Participated International Press, SUMUN 2025, Sarvajanik University
- Volunteer — Cyber Workshop, Sarvajanik University, Surat
- Volunteer — SAW 3.0 2025, Pre-Symposium, SCET